Mannepula Rishith

co24btech11014@iith.ac.in | mannepularishith@gmail.com | +91 8143608178 | LinkedIn | GitHub

EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2028	6.78
Sri Chaitanya Junior College	Class XII (Telangaana Board of Intermediate Education)	2024	93.60%
Sri Chaitanya EM High School	Class X (Board of Secondary Education)	2022	97.00%

POSITIONS OF RESPONSIBILITY

Core Member - Dance Club, IIT Hyderabad

- **Led** and **coordinatedclub activities**, **practice sessions**, **choreography**, and **event planning**, while **participating** in **various dance competitions**.

PROJECTS

Hybrid Movie Recommendation System

Python, Flask, PostgreSQL, Pandas, NumPy, Scikit-learn, Git

- Developed a **Hybrid Movie Recommendation System** by combining **Content-Based Filtering** and **Collaborative Filtering** to generate personalized movie recommendations.
- Built a **Flask-based web application** enabling users to search movies, receive real-time recommendations, and interact through a responsive user interface.
- Designed and managed a **PostgreSQL database** to store movie metadata, user profiles, ratings, watch history, and recommendation-related data.
- Performed **data cleaning, preprocessing, feature engineering**, and exploratory analysis using **Pandas** and **NumPy**.
- Implemented **TF-IDF Vectorization**, **Cosine Similarity**, and **K-Nearest Neighbors (KNN)** using **Scikit-learn** to compute content and user-item similarities.
- Integrated external movie metadata and poster information using REST APIs to enrich user recommendations.
- Optimized recommendation retrieval through efficient **dictionary-based lookups**, **hash maps**, and **priority queues (Top-K recommendations)** for faster response times.
- Version-controlled the project using **Git** and documented the architecture, setup, and deployment process in a comprehensive README.

E-Commerce Analytics Platform *(In Progress)*

- Developed a **Flask-based E-Commerce Analytics Platform** to analyze customer behavior, product performance, and sales trends using interactive dashboards.
- Wrote **complex SQL queries** involving **JOINs**, **CTEs**, **Window Functions**, **Aggregations**, and **Indexes** to generate business insights efficiently.
- Implemented **RFM (Recency, Frequency, Monetary) Analysis** for customer segmentation and purchasing behavior analysis.
- Developed **sales forecasting models** and business KPIs to support data-driven decision-making.
- Used **SQLAlchemy** for database integration and optimized database queries to improve dashboard performance.

SKILLS

Programming Languages : Python, SQL, C/C++

Libraries/Frameworks : Pandas, NumPy, Matplotlib, Scikit-learn, SQLAlchemy, Flask

Databases : PostgreSQL, MySQL, SQLite

Tools : Git, GitHub, Jupyter Notebook, VS Code, Linux (Ubuntu)

RELEVANT COURSES

Computer Science : Introduction to Programming, Data Structures and Applications, Operating Systems

Mathematics : Probability, Statistics, Linear Algebra, Differential Equations

Scientific Computing : Fundamentals of Scientific Computing, Computer Aided Numerical Methods, Computational Methods in Material Science

EXTRACURRICULAR

Coordinated with a **25-member team** and **participated** in the **Group Dance Event** at the **Inter IIT Cultural Meet 8.0**, **IIT Kanpur**.

Volunteered in multiple community service drives organized by the **National Service Scheme (NSS)**.

Participated in a **Zonal-Level Tennis Tournament**.